This is a hands-on manual for creating the "JetBot Image" from the scratch. This guide uses a Jetson Nano Developer Kit 4GB variant.

Prerequisite:

1. Jetson Nano
2. SD card (at least 32GB, but 64GB or higher is recommended)
3. Card Reader
4. A windows laptop or desktop
5. Balena Etcher or similar software package to flash the SD card.
6. A 5V 2A micro-USB power cable.
7. HDMI cable, keyboard and mouse, and a monitor or a laptop or desktop with Ubuntu for headless mode.

Step-1: Flashing the SD card

1. Download the standard JetPack image from this link and extract the file in any preferred location.
2. Insert the SD card in the card reader and format it using Windows 10 or 11's in-built format option or download, install, and launch SD Memory Card Formatter for Windows.
3. Download, install, and launch Etcher.
4. Open Etcher, select the SD card image downloaded in step-1, select the drive, and click flash. Depending on the read/write rate of the SD card, this process can take 10-60 minutes.
5. Once done, there will be action box asking to format the drive. Just click "Cancel" for each one and eject any of the drives. The SD card is now flashed with the standard JetPack image.

Step-2: Preparing the Physical setup and first boot

Follow this link for this step.

Step-3: Enabling i2c

1. Open Terminal
2. sudo usermod -aG i2c $USER
3. after it's done, restart jetson nano

Step-4: Install pip and some python dependencies

1. sudo apt-get update
2. sudo apt upgrade
3. sudo apt install nano
4. sudo apt install python3-pip python3-pil
5. sudo pip3 install --upgrade numpy

Step-5: Installing Tensorflow pip wheel

1. sudo apt-get update
2. sudo apt-get upgrade -y

3. sudo apt-get install -y python3-pip python3-dev python3-numpy libhdf5-serial-dev hdf5-tools libhdf5-dev zlib1g-dev zip libjpeg8-dev liblapack-dev libblas-dev gfortran
4. sudo pip3 install -U pip==20.3 setuptools==65.5.0
5. sudo pip3 install 'protobuf<4' 'Cython<3'
6. wget --no-check-certificate \
7. https://developer.download.nvidia.com/compute/redist/jp/v461/tensorflow/tensorflow-2.7.0+nv22.1-cp36-cp36m-linux_aarch64.whl
8. sudo pip3 install tensorflow-2.7.0+nv22.1-cp36-cp36m-linux_aarch64.whl

Step-6: Installing PyTorch and Torchvision

For PyTorch

1. wget https://nvidia.box.com/shared/static/p57jwntv436lfrd78inwl7iml6p13fzh.whl -O torch-1.8.0-cp36-cp36m-linux_aarch64.whl (replace 1.8.0 with whatever PyTorch version desired)
2. sudo apt-get install python3-pip libopenblas-base libopenmpi-dev libomp-dev
3. pip3 install 'Cython<3'
4. pip3 install numpy torch-1.8.0-cp36-cp36m-linux_aarch64.whl (replace 1.8.0 with the version downloaded)

For Torchvision

1. sudo apt-get install libjpeg-dev zlib1g-dev libpython3-dev libopenblas-dev libavcodec-dev libavformat-dev libswscale-dev
2. git clone --branch v0.9.0 https://github.com/pytorch/vision torchvision (replace v0.9.0 with the torchvision for the PyTorch downloaded earlier)
3. cd torchvision
4. export BUILD_VERSION=0.9.0 (replace 0.9.0 with the version downloaded using the immediate previous commandline)
5. python3 setup.py install –user
6. cd ../

Step-7: Installing Traitlets

1. sudo apt update
2. sudo pip3 install traitlets

Step-8: Installing Jupyter lab

Installing nodejs

1. sudo apt update
2. sudo apt install curl
3. curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.3/install.sh | bash
4. close and reopen the terminal
5. nvm install xx.xx.x (replace xx.xx.x with whatever Node.Js you want)
6. nvm use xx.xx.x

Installing Jupyter Lab

1. sudo pip3 install jupyter jupyterlab
2. sudo jupyter labextension install @jupyter-widgets/jupyterlab-manager
3. jupyter lab --generate-config
4. jupyter notebook password
5. Use any preferred password

Step-9: Installing Repo

1. Sudo apt install python3-smbus
2. git clone https://github.com/NVIDIA-AI-IOT/jetbot
3. cd jetbot
4. sudo apt-get install cmake
5. sudo python3 setup.py install

Step-10: Installing JetBot services

1. cd ~/jetbot/jetbot/utils
2. python3 create_stats_service.py
3. sudo mv jetbot_stats.service /etc/systemd/system/jetbot_stats.service
4. sudo systemctl enable jetbot_stats
5. sudo systemctl start jetbot_stats
6. nano create_jupyter_service.py and use this python script

```
[Unit]
Description=JetBot Jupyter Notebook
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
# Replace 'your_username' below with your system user (e.g. 'edgeai')
ExecStart=/home/your_username/.local/bin/jupyter notebook --NotebookApp.token=''
--ip=0.0.0.0 --port=8888 --notebook-dir=/home/your_username/jetbot --no-browser
WorkingDirectory=/home/your_username/jetbot
User=your_username
Group=your_username
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
```

7. python3 create_jupyter_service.py
8. sudo mv jetbot_jupyter.service /etc/systemd/system/jetbot_jupyter.service
9. sudo systemctl enable jetbot_jupyter
10. sudo systemctl start jetbot_jupyter

Step-11: Make swapfile

1. sudo fallocate -l 4G /var/swapfile
2. sudo chmod 600 /var/swapfile
3. sudo mkswap /var/swapfile
4. sudo swapon /var/swapfile
5. sudo bash -c 'echo "/var/swapfile swap swap defaults 0 0" >> /etc/fstab'

Step-12: Copy JetBot notebooks to home directory

cp -r ~/jetbot/notebooks ~/Notebooks